

Human & Computer Interaction

week 1

Aj.Pattama Charoenporn

Question

“ 1. Why this course is important for a software developer?”

“2. What do you expect to learn from this course”

Week 1 - Greeting Agenda



- Course Description
- The goals of this course
- Course evaluation
- Course Schedule
- An Introduction of HCI

The Rules for this course

- เริ่ม 9.30 -12.00
- ไม่มีเช็คชื่อเข้าห้อง
- ตั้งชื่องานส่ง -> รหัสนศ. – ชื่อ นามสกุล
- ไม่จับกลุ่มทำงานข้าม section

แหล่งความรู้เพิ่มเติม



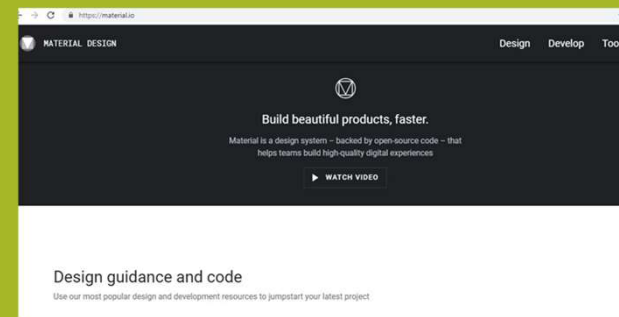
Interaction-Design.org ✓ Liked ▼
394K like this · Non-profit organisation
Learn about UX Design through our open-source educational mate...



UX Thailand Meetup ✓ Liked ▼
Page
Community
1.8K like this



UI/UX designers + Join
84K members · 10+ posts a day
This group is meant for designers - a place to learn and share - to ask questions, network, and improve....



Course Description

ความรู้เบื้องต้นเกี่ยวกับปฏิสัมพันธ์ระหว่างมนุษย์และคอมพิวเตอร์
ความสามารถของประสาทสัมผัสของมนุษย์ ระบบติดต่อกับผู้ใช้แบบ
กราฟิกส์ การพัฒนาระบบโต้ตอบ การออกแบบส่วนปฏิสัมพันธ์ระหว่าง
มนุษย์และคอมพิวเตอร์สำหรับระบบสารสนเทศแบบมัลติมีเดีย รวมถึง
เทคโนโลยีและเครื่องมือใหม่ ๆ ทางด้านคอมพิวเตอร์กราฟิกส์ และการ
นำไปใช้งาน

Course Evaluation

Single work	15%
Group work (5 people/group)	35%
Participate + Quiz	10%
Midterm	40 %

Course Schedule:

Midterm

Week 1

Greeting & Introduction



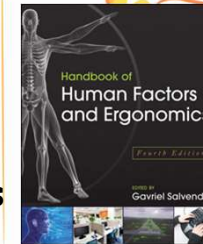
Week 2

Human perception



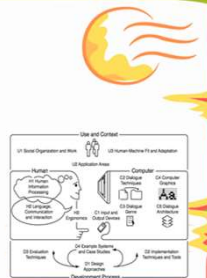
Week 3

Human factors & Ergonomics



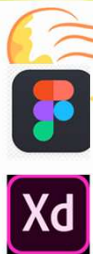
Week 4

Interaction Model



Week 5

Introduction to UI design tools



Week 6

The Principles of GUI



Week 7

User Experience (UX) + Consumer Behaviour



Week 8

Midterm Exam



Course Schedule: Final

Week 9

Website
Design



Week

10
Mobile App
Design



Week 11

Project work



**Week
12**

Project
work



**Week
13**

Project
work



Week

14
Project
presentation



**Week
15**

Review+
Submit
Singlework



**Week
16**

Review+
Submit
Singlework



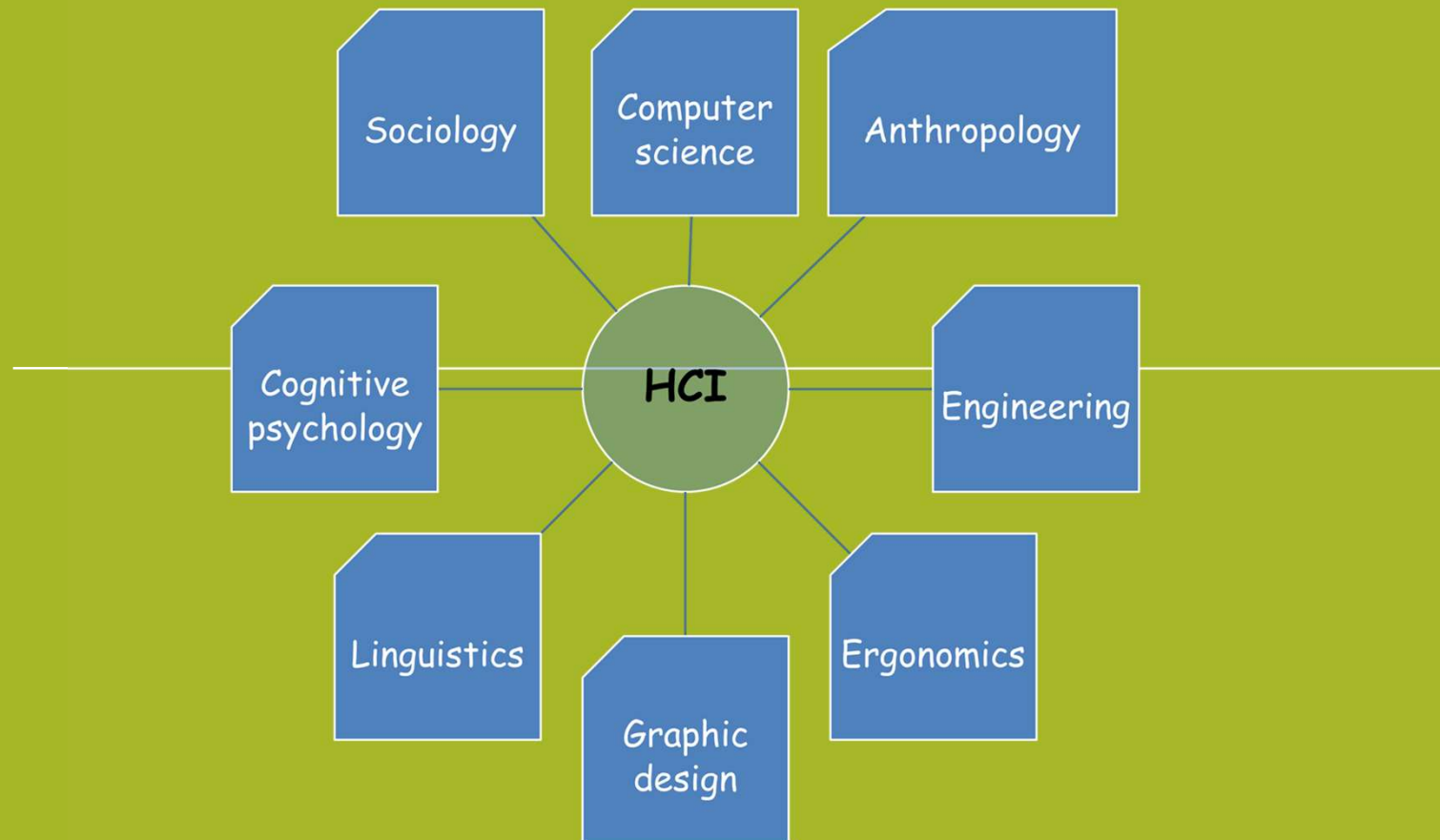
An Introduction of HCI



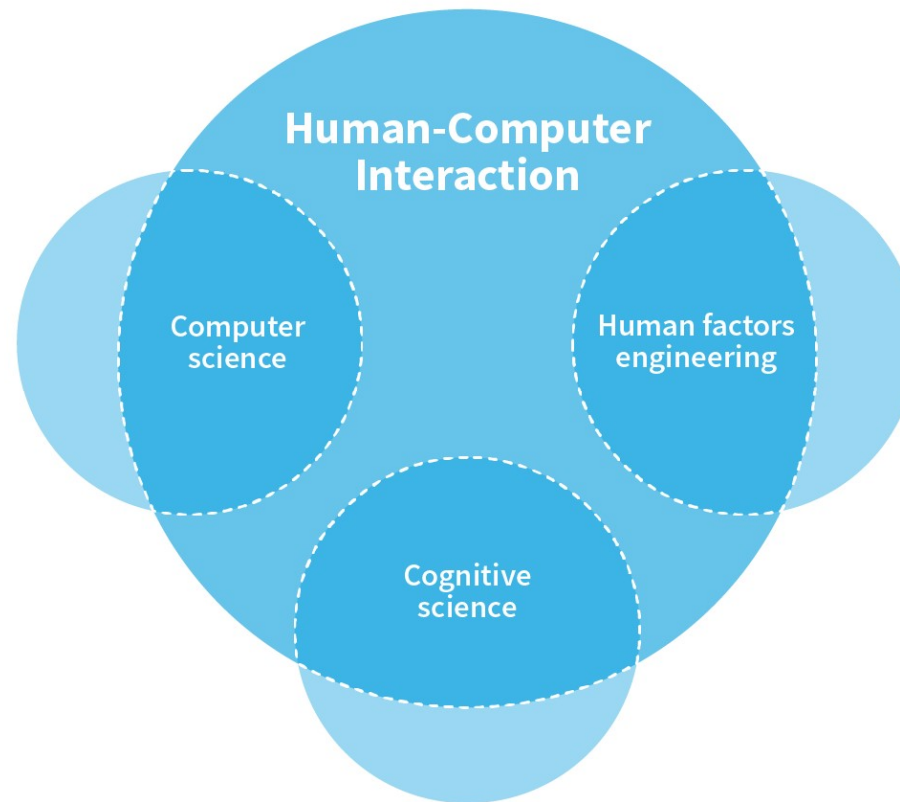
"Human-computer interaction is a discipline concerned with the design, evaluation, and implementation of interactive computing systems for human use and with the study of major phenomena surrounding them."

Thomas T. Hewett et al., 1992

Human-computer interaction and related fields

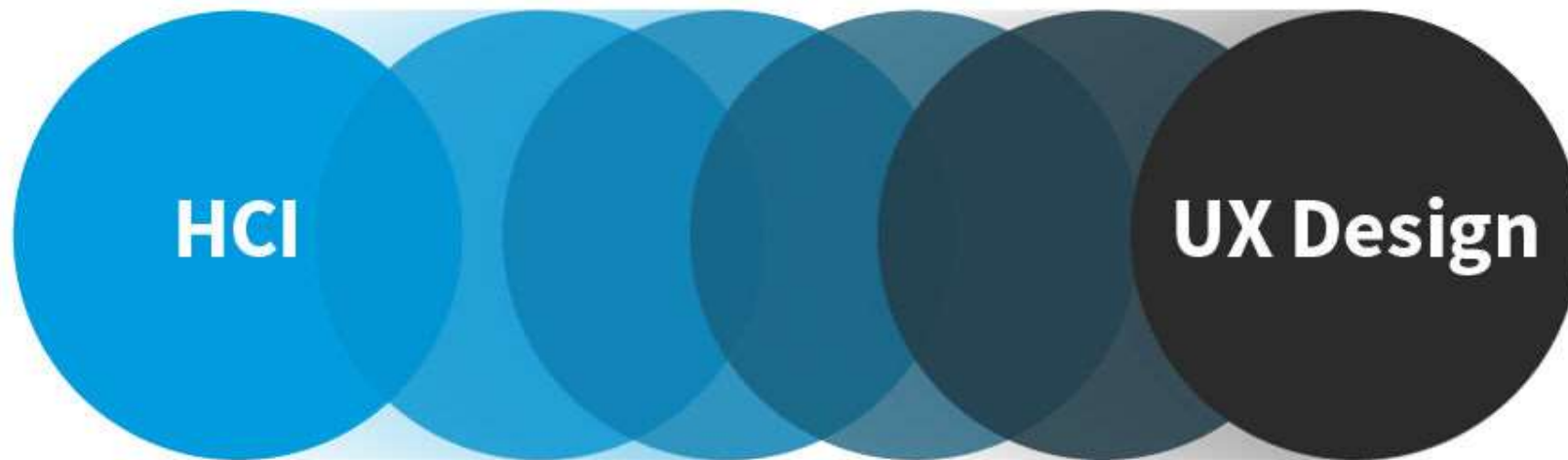


The Multidisciplinary Field of HCI

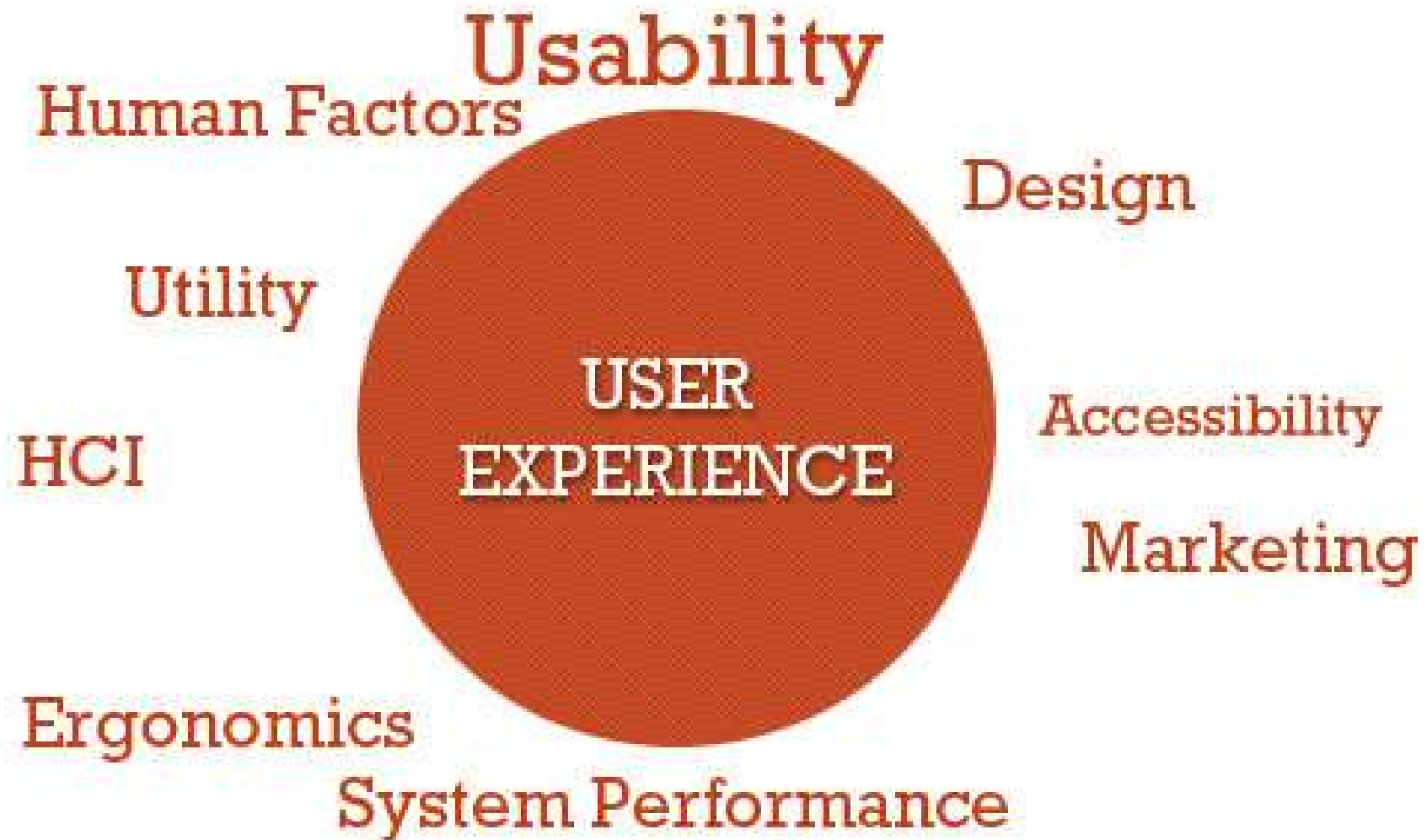


Interaction Design Foundation
interaction-design.org

The Evolution of UX Design



Interaction Design Foundation
interaction-design.org



UX

UI



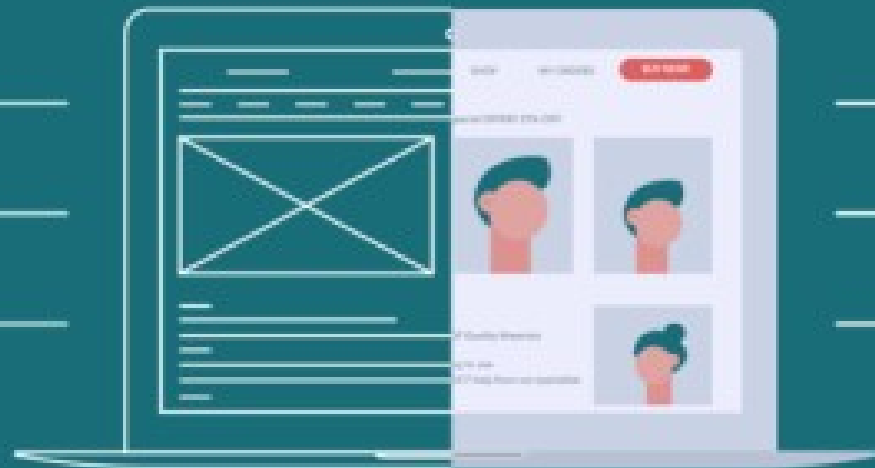
User
research



Personas



Usability
testing



Interaction
design

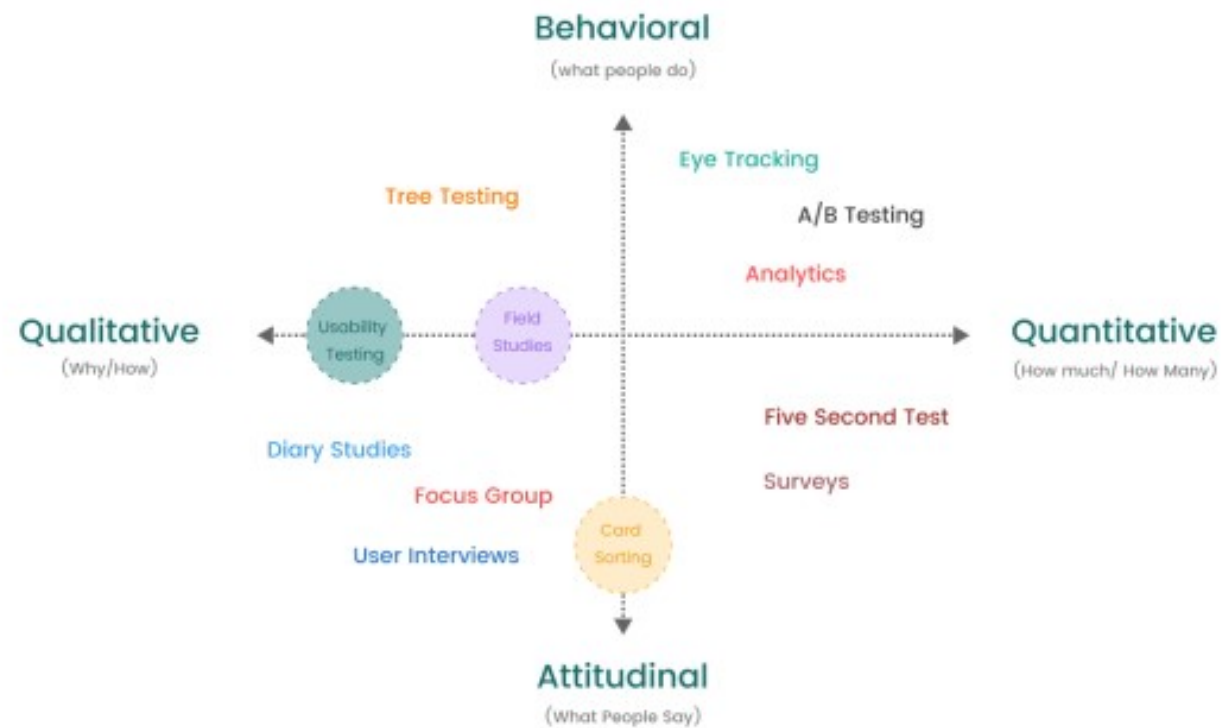


Layout



Visual
design

User Research Methods



The history of HCI

1950s - computers invented

1960s - first screen and mouse developed

1980s - Arrival of microchip and micro-computers

1984 - Apple Macintosh (Xerox Star). First conference in HCI.

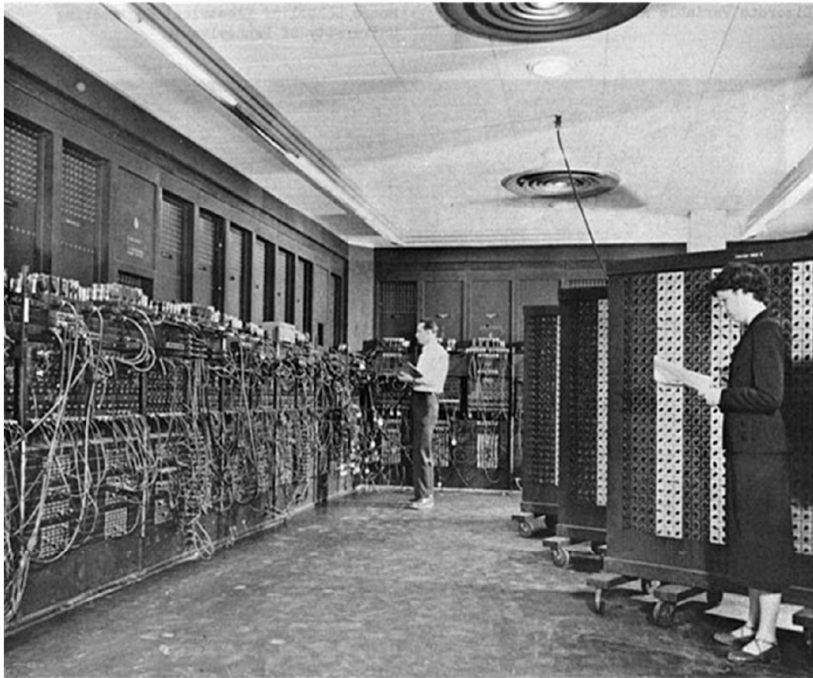
1970s - Robots

1990s - World Wide Web arrives: Websites & Smartphones

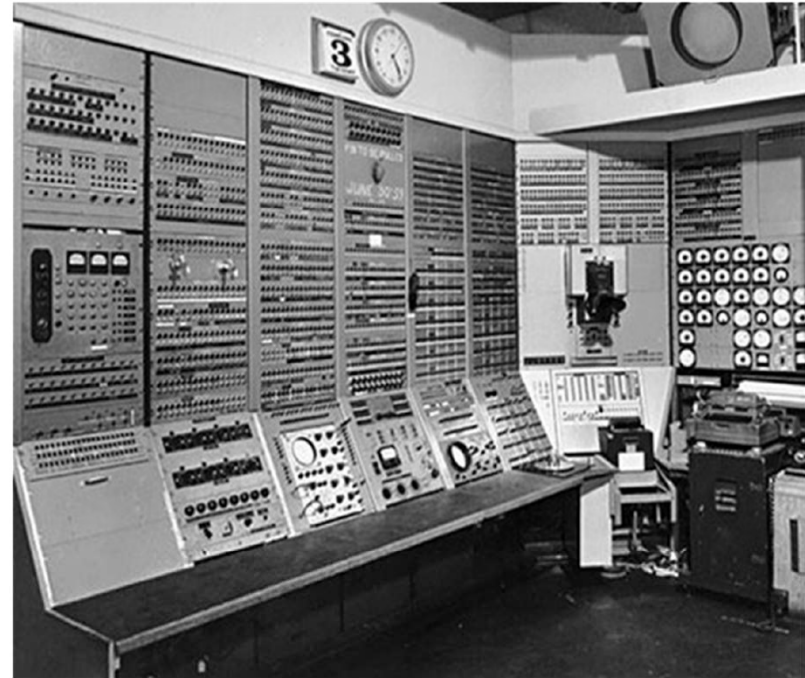
2000s- AR/VR technologies

First Computer

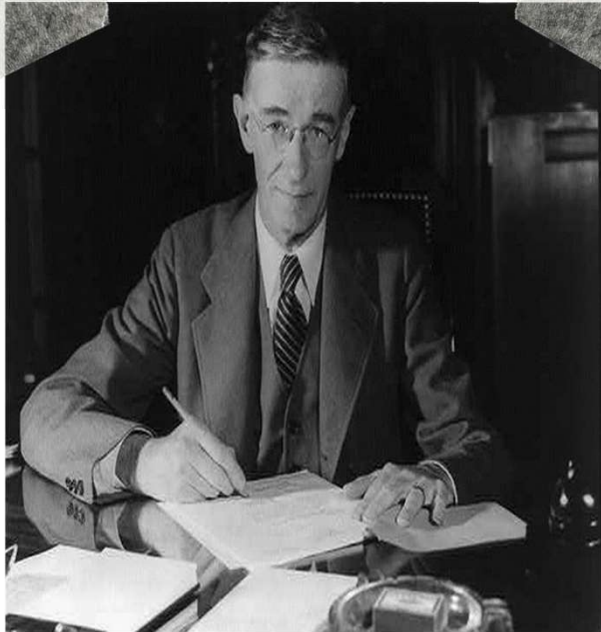
The first digital computer - 1937



The first computer with RAM - 1955



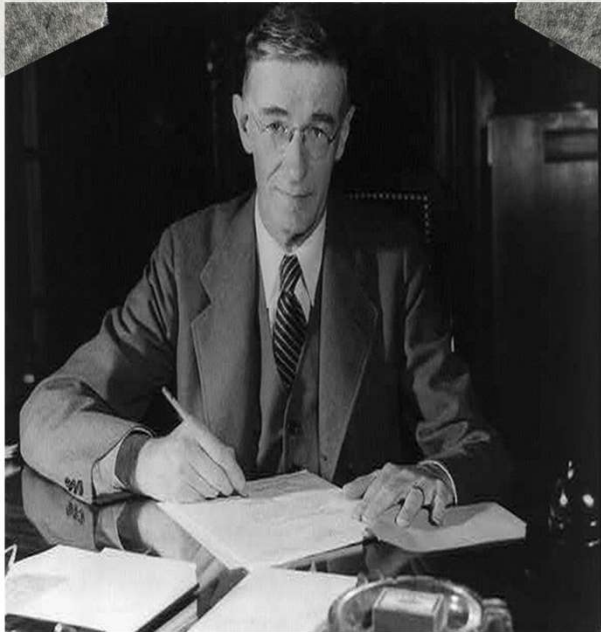
Vannevar Bush



- He was an American engineer, inventor and science administrator
- He published "As We May Think" - 1945 Atlantic Monthly
- He Developed Memex
 - Stores all records/articles/communications
 - Items retrieved by indexing, keywords, cross references (now called hyperlinks)
 - (Envisioned as microfilm, not computer)



Vannevar Bush



Ivan Sutherland



- He is considered to be the father of computer graphics using SketchPad .
- Sketchpad was a computer program written by Ivan Sutherland in 1963 in the course of his PhD thesis.
- Sketchpad was influenced by the conceptual Memex of Vannevar Bush
- It pioneered the way for human-computer interaction (HCI)

Ivan Sutherland



Douglas Engelbart



What is in his hand?

Students, write your response!

Douglas Engelbart



- He invented a mouse.
- It was called "X-Y position indicator for a display system"
- It makes people to interact with a computer directly and easily.
- It was first used with Xerox Alto in 1973.



Mouses

From Computer Desktop Encyclopedia
Reproduced with permission.
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Brian Tramontana, Photographer



Xerox Star



- It is the first personal computer using for "business professionals"
- It provides the concept of **WYSIWYG**.
- Developed GUI for easy to use.
- The cost was about \$75,000



- [illegible]

1983: Birth of HCI

In 1969 ACM's Special Interest Group on Social and Behavioural Computing (SIGSOC) was formed.

The first ACM SIGCHI conference (1983) - Special Interest Group on Computer-Human Interaction.

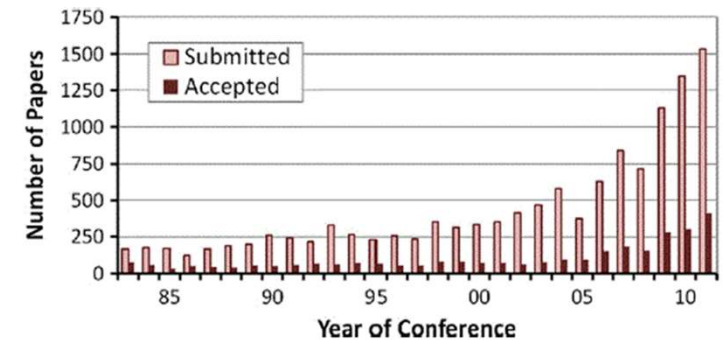


FIGURE 1.9

Number of papers submitted and accepted by year for the ACM SIGCHI Conference on Human Factors in Computing Systems ("CHI"). Statistics from the ACM Digital Library.

1983: The psychology of human-computer interaction

The Psychology of Human-Computer Interaction

STUART K. CARD
THOMAS P. MORAN
ALLEN NEWELL



Emerged from work done at Xerox PARC
Card and Moran arrived at PARC in 1974
Newell was a consultant to the project.

This book provides basic psychology involving in
the human sensory, cognition, and motor output.

HCI in the past

- Focused on personal behavior
- Focused on creating Software
- Research in HCI was done mainly for Desktop



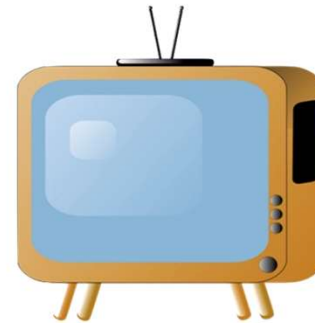
HCI in the present

- Focused on social behavior
- Focused on different behavior in different users' group
- Research was done in different platforms



User experience

Interactive technologies



Clipart Of 1212423



AR & VR Technology

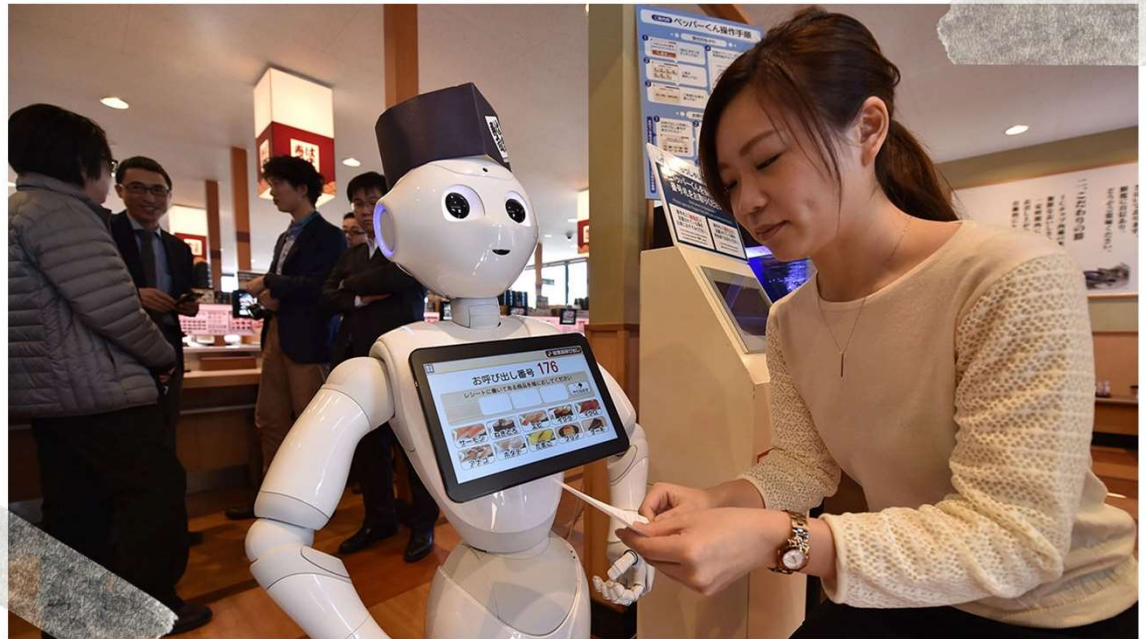


Virtual Reality (VR)



Augmented Reality (AR)

The Humanoid AI Robots

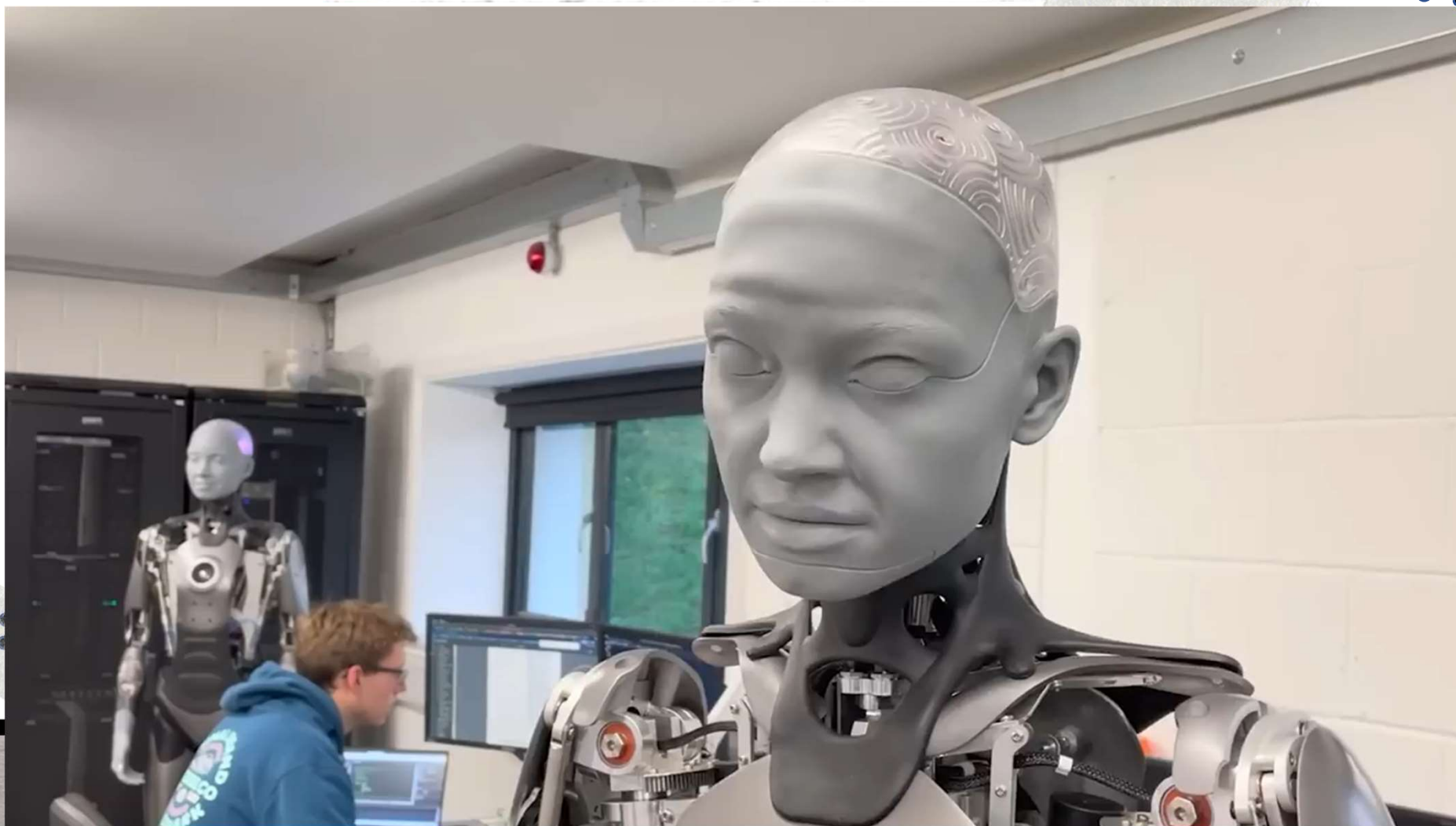


<https://www.thairath.co.th/news/tech/851606>



HOME

The Humanoid AI Robots



single work

จงออกแบบ mobile application เพื่อให้ความรู้เกี่ยวกับเรื่องอะไรก็ได้ โดยกำหนดกลุ่มผู้ใช้งานให้ชัดเจนและออกแบบเพื่อกลุ่มผู้ใช้งานนั้นๆ โดยต้องออกแบบหน้า Homepage และอีก 1 page ที่ลิงก์ไปได้ แต่หน้าต้องมี interactive (มีปุ่มกด) ได้ ภาษาไทยหรืออังกฤษก็ได้แต่ต้องเขียนให้ถูกต้อง และต้องโชว์ทักษะการใช้เครื่องมือการออกแบบและโชว์ความสวยงาม ความคิดสร้างสรรค์ให้อาจารย์เห็น

ห้ามทำงานกลุ่มข้าม Sec!

Team work

5 people: Design a Website Mockup using FIGMA to create your mockup.

- Mockup
- design system
- Prototype
- Main frame
- Basic Feature = register, sign-in, sign-out
- Killing Feature = 1 Feature ที่มาเพื่อฆ่าแอปอื่น และสุกผู้ใช้งานได้อย่างอยู่หมัดเคล็ดลับที่ส่งแอปไปสู่ดวงดาว



Keywords

UI

Ergonomics

HCI

Agile
Vs
Waterfall

Human
Perception

UX

Usability 

Metaphor 

Affordance 

Navigation 

Next week

Week 2

Human perception

